

2.1

India skilling landscape

As per the publication World Economic Forum's "Future of Jobs Report 2023", analytical thinking is the core skill for mass employment which ranks first, creative thinking, ranks second, ahead of three self-efficacy skills - resilience, flexibility and agility, motivation and self-awareness. AI and big data skills are gaining precedence and is a most strongly prioritized skill in certain industries.

The India Economic Survey 2023-24 reveals that Indian economy needs to generate an average of nearly 78.5 lakh jobs annually until 2030 in the non-farm sector to cater to the rising workforce presenting both opportunities and challenges.

Government initiatives:

The National Education Policy (NEP)¹⁸ 2020 has given guidance for revision and revamping of all aspects of education, including the educational structure, regulations, and governance, to create a new system which is aligned with the aspirational goals of 21st century students. According to the policy, by 2025, at least 50% of learners through the school and higher education system shall have exposure to vocational education, for which a clear action plan with targets and timelines are to be developed.

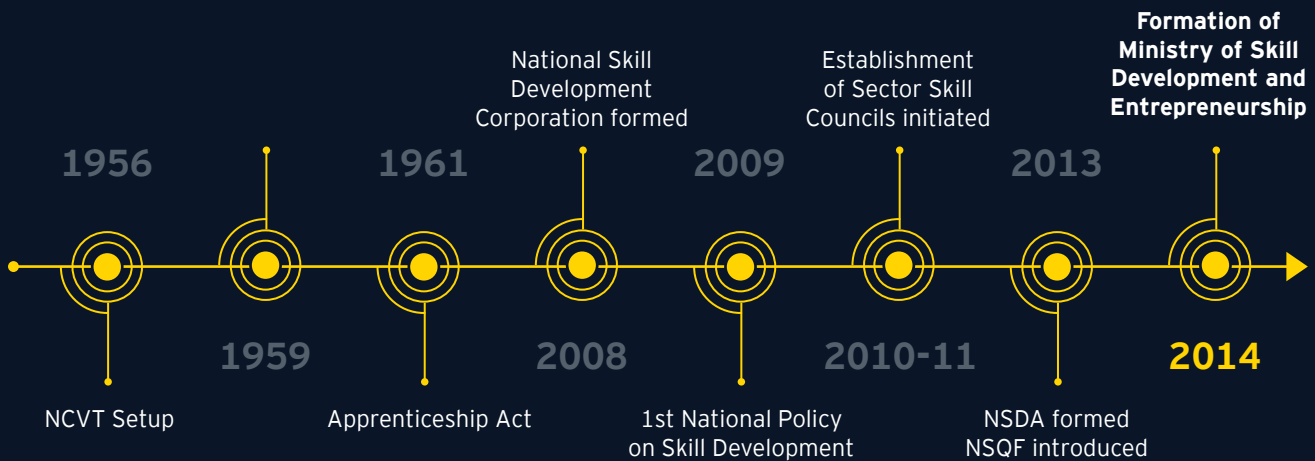
The Ministry of Skill Development and Entrepreneurship (MSDE) is responsible for co-ordination of all the skill development efforts across the country, removal of disconnect between demand and supply of skilled

manpower, building the vocational and technical training framework, skill up-gradation, building of new skills and innovative thinking not only for existing jobs but also jobs that are to be created. It is aided in these initiatives by its functional arms - Directorate General of Training (DGT), National Skill Development Agency (NSDA), National Council for Vocational Education and Training (NCVT), National Skill Development Corporation (NSDC), National Skill Development Fund (NSDF) and 38 Sector Skill Councils (SSCs) as well as 33 National Skill Training Institutes (NSTIs/NSTI(w)), about 15000 Industrial Training Institutes (ITIs) under DGT and 290 training partners registered with NSDC.

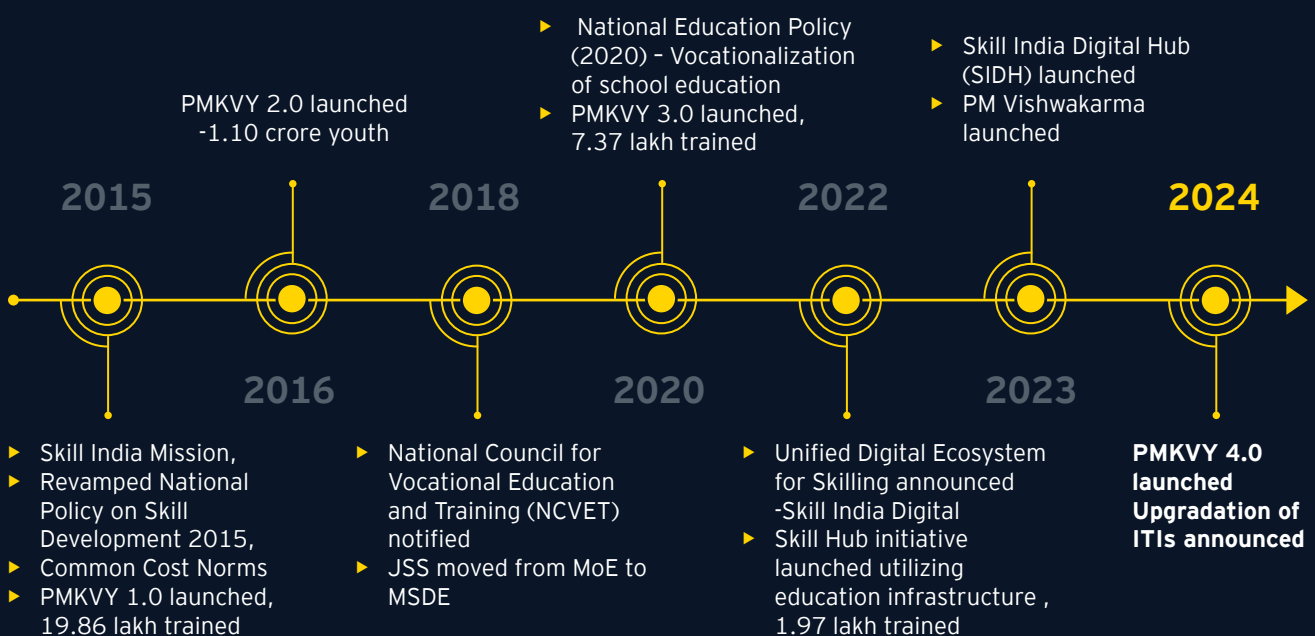
The following landscaping illustration outlines the key interventions launched by the central government towards improving youth skilling and development over the years.

Journey of Indian skilling ecosystem

Evolution journey of competency based skill development ecosystem in India

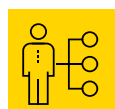


Journey of Skill Development: Post Ministry formation



2.2

Overview of key schemes



Pradhan Mantri Kaushal Vikas Yojana

Pradhan Mantri Kaushal Vikas Yojana - PMKVY was introduced first in 2015. The scheme promotes skill development among young people across the country through provision of free and short trainings along with recognition of informally gained skills (through skilling, reskilling and upskilling). The scheme has undergone various iterations to ensure that it remains relevant, and its delivery stays efficient¹⁹.

The scheme relies heavily on State Governments for implementation and monitoring processes. With PMKVY 2.0, 25% of the overall implementation target was dependent entirely on State Governments through State Skill Development Missions. Under this scheme 812 Pradhan Mantri Kaushal Kendras (model training centres) have been introduced across 707 districts and 540 Parliamentary Constituencies²⁰ as of December 2020. From the first and second edition of the PMVKY scheme, more than 12.6 million young people have been trained successfully⁷.

Under the three phases of PMKVY (i.e., PMKVY 1.0, 2.0, 3.0) a total of 1.37 crores candidates have been trained/ oriented and 1.10 crores are certified across the country. Total reported placements of (under STT and Special Projects) is 24.36 lakh candidates till 2022.

PMKVY 4.0 announced in 2023 with emphasis on emphasize on on-job training, industry partnership, and alignment of courses with needs of industry. She further highlighted that the scheme will also cover new age courses for Industry 4.0 like coding, AI, robotics, mechatronics, IOT, 3D printing, drones and soft skills.



PM Vishwakarma

PM Vishwakarma Scheme launched in 2023 to enable the recognition of artisans and craftspeople as Vishwakarma, making them eligible to avail benefits under the scheme. The scheme aims to support artisans and craftspeople who work with their hands and tools, including 18 trades such as carpenters, blacksmiths and potters. It has provision for skill upgradation to hone their skills, provide support for better and modern tools to enhance their capability, productivity, and quality of products and services, provide the beneficiaries an easy access to collateral free credit and reduce the cost of credit by providing interest subvention, provide incentives for digital transactions, provide a platform for brand promotion and market linkages to help them access new opportunities. The scheme aims at providing several benefits to the Vishwakarmas, who are either self-employed or intend to setup their own small-scale ventures.

The scheme seeks to promote empowerment of women, and those belonging to the marginalized or under-served groups like the Scheduled Castes, Scheduled Tribes, OBCs, Specially Abled, Transgenders, residents of NER states, Island Territories and Hilly Areas. There are 16,58,000 applicants registered under the scheme currently²¹.



Long-term skill development through Directorate General of Training (DGT)

Long-term skill development through Directorate General of Training (DGT)

- ▶ Craftsmen Training Scheme (CTS)
- ▶ Apprenticeship Training Scheme (ATS)
- ▶ National Apprenticeship Promotion Scheme (NAPS) for Designated Trade
- ▶ Crafts Instructor Training Scheme (CITS)
- ▶ Sub-Mission on Polytechnics under the Coordinated Action for Skill Development
- ▶ Women Training
- ▶ Skills Strengthening for Industrial Value Enhancement (STRIVE)
- ▶ Enhancing Skill Development Infrastructure in North eastern States (ESDI)

“

As the Digital Public Infrastructure (DPI) for Skilling, Education, Employment, and Entrepreneurship, SIDH serves as a comprehensive information gateway for government initiatives in these domains, making it a go-to hub for citizens pursuing career advancement and lifelong learning

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Shri Jayant Chaudhary,
Minister of State
(Independent Charge) Ministry
of Skill Development, and
Entrepreneurship
*in a written reply in the Rajya Sabha*²³



National Apprenticeship Promotion Scheme

National Apprenticeship Promotion Scheme - NAPS is a flagship program under the Ministry of Skill Development and Entrepreneurship (MSDE) designed to promote apprenticeship training in India. It offers structured, industry-aligned training to young people, boosting their employability and contributing to the nation's skilled workforce. NAPS fosters collaborations between industries and training providers to ensure practical learning and relevant skill development. According to the Apprenticeship India report for 2023, there are 931,988 apprentices currently in training, with 500,805 having completed their training and 35,387 certified. In 2023, claims totalling INR43 crore have been disbursed. For 2024, approximately 324,800 apprentices are engaged in training, with INR395 crore in Direct Benefit Transfer available across India and 634,294 live vacancies²².



Skill India Digital Hub - SIDH

Skill India Digital Hub - SIDH is a comprehensive digital platform designed to synergize and transform the skills, education, employment and entrepreneurship landscape in India. SIDH plays a crucial role in preparing the Indian workforce for Industry 4.0 by offering futuristic courses on areas such as, Big Data, Machine Learning and Analytics, provided by its digital learning partners. The hub also offers several Industry 4.0 courses such as:

- ▶ Python with Advanced Artificial Intelligence (AI)
- ▶ Artificial Intelligence Foundation
- ▶ Generative AI
- ▶ Build Classical Machine Learning Models with Supervised Learning
- ▶ Data Analytics Essentials
- ▶ Analytics Data in a Relational Data Warehouse
- ▶ Cyber security Essentials
- ▶ Introduction to Data Science
- ▶ Kisan Drone Operator
- ▶ EV Service Technician
- ▶ Bio-waste Management
- ▶ Other certification courses



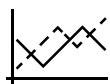
Prominent skilling initiatives taken up by other ministries include -

- ▶ **Ministry of Labour and Employment (MoLE)'s** eSHRAM portal for creating a National Database of Unorganized Workers (NDUW) and eMigrate for the Overseas Employment Division of Ministry of External Affairs (MEA).
- ▶ **Ministry of Rural Development (MoRD)'s Deen Dayal Upadhyaya Grameen Kaushal Yojana (DDU-GKY)** - was launched in September 2014. The scheme focuses on diversifying family income of the rural poor and providing skilling and career support. The scheme emphasizes providing support to youth (15- 35 years old) in rural areas. Through this scheme, young people who have been unable to complete their education are able to obtain vocational training to build their careers. As of 2023, a total of 14.51 lakh candidates have been trained and 8.70 lakh candidates placed under the program. Further, a sum of INR7015.61 crore was released under DDU-GKY in 2023.
- ▶ **National Urban Livelihoods Mission** - NULM focuses on providing educational and skilling opportunities specifically to the urban poor. The main aim of this scheme is to reduce poverty and vulnerability of the urban poor by facilitating their access to different employment opportunities (including self-employment, wage employment, etc.), and access to shelter for those without homes and providing social security. According to Annual report of NULM 2023, 1,92,122 candidates were skilled and trained, while 89,432 candidates were placed in jobs.
- ▶ **Ministry of Agriculture and Farmers' welfare scheme** - Agri-Clinic Agri-Business centres scheme (ACABC) with NABARD acting as subsidy channelizing agency.
- ▶ **Ministry of Textiles scheme Integrated Skill Development Scheme (ISDS)** to upgrade the skills of textile workers/ handloom weavers and Samarth (Scheme for Capacity Building in Textiles Sector).
- ▶ **National Apprenticeship Training Scheme** - NATS is a flagship scheme under the Ministry of Skill Development and Entrepreneurship (MSDE) aimed at promoting apprenticeship training in India. It provides structured, industry-aligned training to young people, enhancing their employability and contributing to the country's skilled workforce. NATS fosters partnerships between industries and training providers to ensure practical learning experiences and relevant skill development. As per the NATS report till date, 36,58,130 students registered, 2,61,703 students undergoing apprentices training. Direct Benefit Transfer (DBT) till date is around INR76.96 crore²⁴.

Even though there is a significant push towards prioritizing critical sectors such as employment, skilling, MSME, the skilling landscape of India continues to face barriers.

**Vocational education and skilling are not aspirational**

India's youth carry negative social perception towards vocational training as they are not considered a viable alternative to traditional formal education. It finds limited acceptance as dignified labor.

**Lack of quality and standards in the skilling space**

India faces scarcity of qualified trainers, quality infrastructure and course content combined with a lack in standardization in assessment and accreditation in vocational education.

**Skill mismatch in the labor market**

Mismatch in educational qualification and skills; often poorly connected to the qualification needed to perform a job.

**Inadequate industry interface**

India lacks alignment between industries and educational and training courses in terms of the demand and supply of skills, lack of industry exposure among students, poor placements and internship opportunities.

**Duplication of efforts due to overlapping skilling initiatives across multiple departments**

A number of government ministries and departments at the central and state level offer multiple skilling schemes and programs which lead to overlap and duplication of efforts and resources.



Private sector initiatives

Numerous private sector organizations are working towards supporting youth for skilling and employment. In addition to collaborating with the NSDC (through Sector Skill Councils), the private sector is invested in the youth through the following strategic engagements:

- Vocational training and capacity building for specific industries
- Working with ITIs, universities and academic institutions for placement drives, hiring of apprentices, on the job training, curriculum and faculty development and much more
- Mentorship and guidance to disadvantaged youth, such as rural youth, urban poor youth, women, youth with disabilities and youth from marginalized communities
- CSR and skilling collaborative initiatives

In a nutshell, India, through its various initiatives and investments is striving towards creating a democratized access to education and training to enable millions of people to acquire the skills they need to succeed

in today's rapidly changing economy. Endorsing its commitment, the recent Indian Union Budget of 2024-2025²⁵, announced several measures for youth and employment, such as:

- ▶ Allocation of INR1.48 lakh crore for education, employment and skilling
- ▶ INR2 lakh crore package announced for five schemes on jobs and skilling of 4.1 crore youth
- ▶ Financial support for loans up to INR10 lakh for higher education in domestic institutions
- ▶ One month wage to all workers newly entering the work cycle. INR15,000 to entrants with monthly salary up to INR1 lakh. This will benefit 210 lakh youth
- ▶ Internship opportunities to one crore youth in 500 top companies across the country in five years
- ▶ Modernization of 1000 Industrial Training Institutes (ITIs) over the next five years through a hub-and-spoke model, supported by a total outlay of INR60,000 crore
- ▶ Priority to be given to increase women's participation in the workforce through the establishment of hostels and partnerships to conduct women-specific skilling programs





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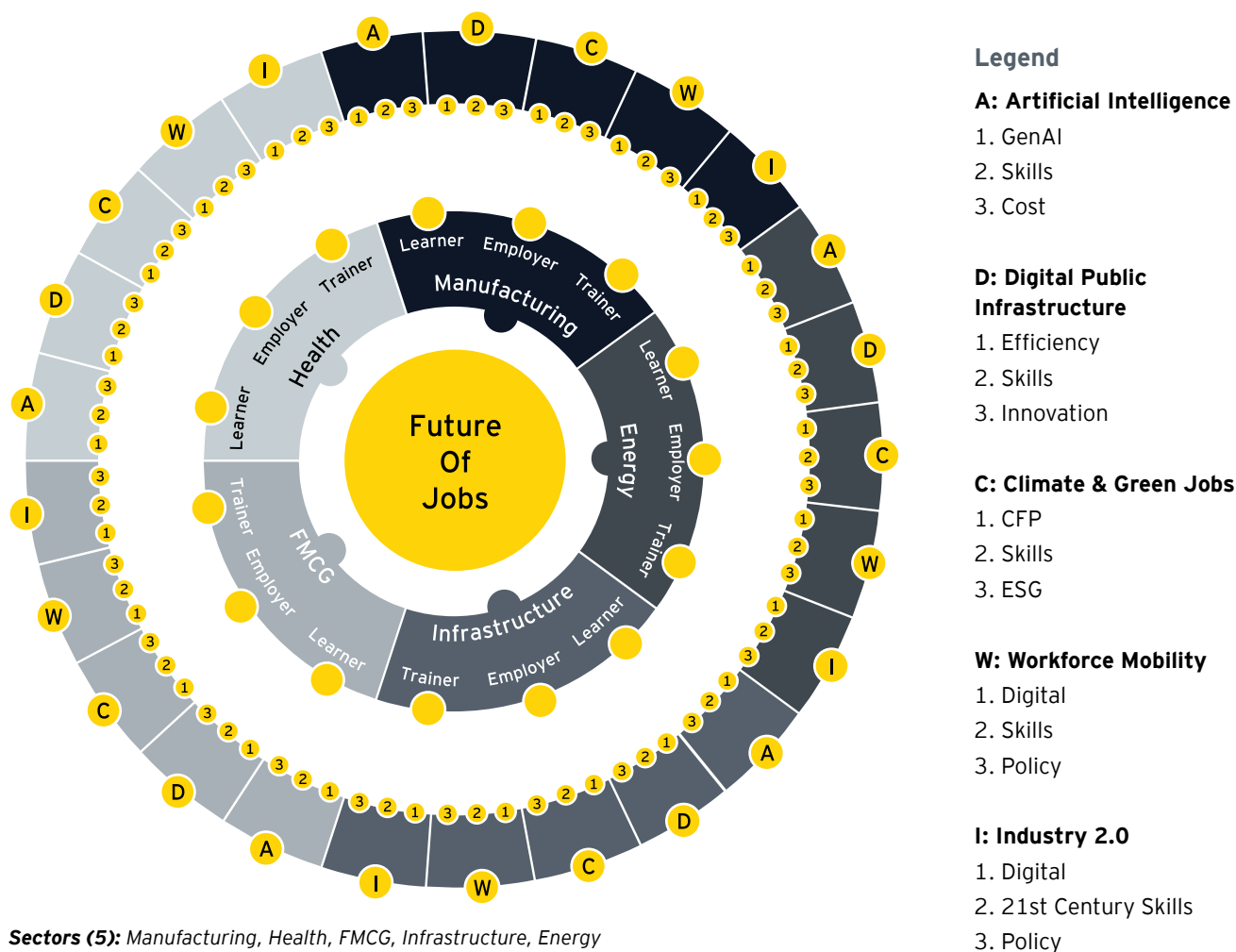
FICCI-EY future of jobs reports

As the job and skilling landscape evolves at a rapid pace with disruptive technologies making way, and COVID-19 pandemic which brought significant changes to the labor market, a committee comprising senior industry experts and panel of FICCI and EY made an informed decision to publish a new edition of FOJ report which studies in-depth the jobs and skills landscape of India across five sectors, namely manufacturing, FMCG, healthcare, energy and infrastructure.

Study methodology

For this report, the committee outlined a framework focused on assessing the impact of five trends– Artificial Intelligence (AI), DPI, Climate, Workforce Mobility and Industry 4.0– on the five sectors and how its interplay will affect three key stakeholders– learners, employers and trainers in the future in terms of jobs and skills.

We further sub-divided the five lenses into key factors, which we then individually considered to assess the impact on all sectors and stakeholders. We completed the assessment with the help of primary (interviews and digital survey) and secondary research (literature review) and then analyzed and drew insights from the data collected.



In the concluding section, we have presented our findings of the assessment in the form of Harvey Balls visualization.

Model of study

In addition to the secondary research, we sought an informed view by interacting with over 200 respondents, a mix of senior leaders and industry experts, learners, trainers, academicians and policy makers. Structured and unstructured form of interview along with digital survey (including multiple choice questions) was part of the methodology. The respondents were probed on their views on industry 4.0 technologies that will set the narrative in the future. They include:

- How the five lenses (AI, DPI, climate, workforce mobility and Industry 4.0 technologies) are expected to impact jobs and skills
- Challenges anticipated with its integration
- Emergence of new job roles or transformation of existing ones if sustainable practices are a priority for business operations and its subsequent impact on green jobs and skills
- Increasing importance of global mobility and cross-cultural skills
- Role of skilling institutions in aligning educational curriculum with the skill requirements of employers
- How effective current government policies are
- Additional support needed from government
- Recommendations to improve the job and skilling ecosystem



Primary research: consultation with stakeholders

- ▶ Identification of industry's top leaders and employers in selected sectors, learners across all levels and trainers
- ▶ Developed questionnaire on digital tool for each of the five sectors
- ▶ Conducted interviews with senior leaders and industry leaders
- ▶ Collation and analyzing of data of primary research



Secondary research: Literature review of future of jobs scenario across the lens of AI/Gen AI, technology, climate change, global mobility, and 21st century skills

- ▶ Comprehensive document review of existing reports, research, statistics, global standards and best practices
- ▶ Quantitative and qualitative research from multiple data sources such as Union Budget 2024-25, Ministry of Skill Development and Entrepreneurship (MSDE), IBEF, Periodic Labour Force Surveys, Economic Survey of 2023-24, ILO and WEF reports
- ▶ Collating and analysing data to identify factors impacting FOJ



Validation by industry leaders

Validation by industry leaders of the findings and recommendations for Learners, Trainers and Employers and the need for Government reforms and support



Below is a summary of previous FOJ reports:

Future of Jobs 1.0: Year 2017

Future of Jobs in India - A 2022 perspective was published by FICCI in 2017 in partnership with NASSCOM and EY. The report was the first research study exclusive to the Indian context on the future of jobs across five key sectors of the Indian economy in detail: IT-BPM, retail, textiles and apparel, automotive and BFSI (Banking, Financial Services, and Insurance).

Future of Jobs 2.0: Year 2018

Future of Jobs in India-2.0 published by FICCI in collaboration with EY in the year 2018 analyzed five more sectors in the second phase of its study. These sectors together form the foundation of India's socio-economic development and are healthcare, education, construction, transportation & logistics and tourism & hospitality. The report was a product of quantitative and qualitative research and strategic foresight received from top leaders and employers in selected sectors for the study.

Future of Work for Women (COVID specific): Year 2021

The report "Impact of COVID-19 and Industry 4.0 on Future of Work for Women" was published by FICCI in May 2021 in collaboration with UNDP. The report studied the unprecedented changes in the world of work necessitated by COVID-19 and how post COVID-19, the situation could change and so will the impact of technology on jobs and employment in India.

Future of Jobs 3.0: Year 2024 (Current report)

This report of Future of Jobs (this report) aims to provide deep insights about job creation, reskilling and upskilling opportunities for the next 2-3 years for learners, trainers and employers. The entire study and research on FOJ 3.0 were conducted over a period of 2.5 months with rigorous effort put in by EY and FICCI.

FOJ 3.0 analyzes five sectors with a lens of emerging technologies, global workforce mobility and provides recommendations for future ready policy. These five sectors are manufacturing, FMCG, healthcare, energy, and infrastructure. The rationale for choice of these five sectors is presented below-